

Validation System for PeerGFS-Based Active-Active NAS DR

Production-Grade Multi-Site DR Validation and Demonstration Plan

Leading Point

2026

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Validation Objectives & Expected Outcomes

Traditional NAS-based DR environments commonly face limitations such as WAN performance degradation, extended RPO windows, repeated retransmission of large files, and underutilized passive DR sites. This validation plan is designed to evaluate whether PeerGFS can support production-scale Active-Active NAS DR environments.

Near Real-Time Active-Active DR Validation

Validation of near real-time data synchronization and bi-directional Active-Active operations

Delta-Level Replication Efficiency Validation

Validation of a changed data transfer architecture designed to minimize network utilization by transmitting only modified data blocks

WAN Environment Validation

Validation of synchronization stability under high-latency and packet-loss network conditions

Failure Recovery and Data Consistency Validation

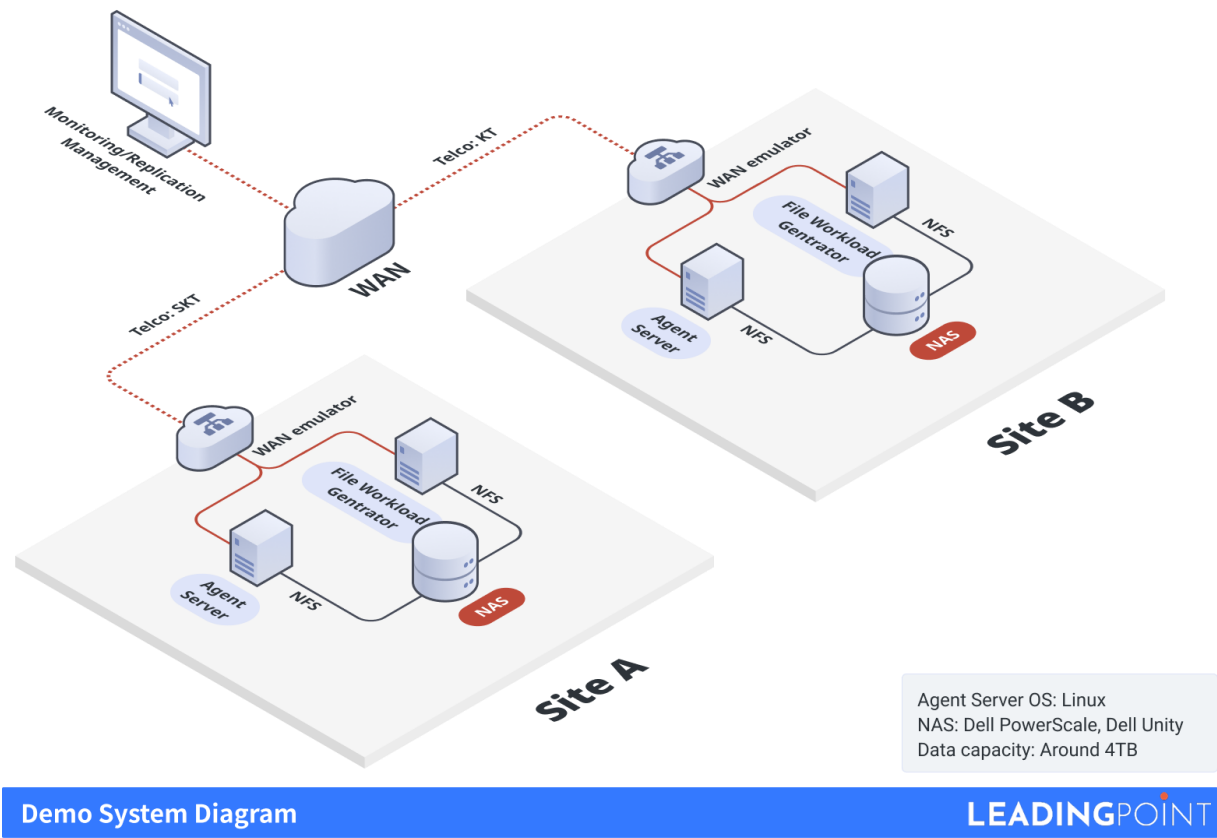
Validation of automatic re-synchronization and data integrity during failure and recovery scenarios

Key Validation Areas

Validation Area	Expected Outcome	Validation Method
Near Real-Time Replication Performance	Measurement of actual replication latency between file modification and DR site synchronization	Timestamp-Based Synchronization Latency Calculation
Delta-Level Replication	Verification of efficient transfer of changed data only	Comparison of transferred data volume after partial file modification
Active-Active File Services	Verification of stable bi-directional operations and data consistency during concurrent workloads	Validation of data consistency after concurrent file creation and modification across both Main and DR sites
WAN Replication Stability	Measurement of replication performance under degraded network conditions	Linux tc-based simulation of latency, packet loss, and bandwidth constraints
Failure Recovery and Data Consistency	Verification of automatic re-synchronization and data consistency after failure recovery	Post-recovery data consistency comparison

Validation Environment and Architecture

The validation environment has been designed to closely emulate a real-world production deployment in order to validate real-time replication, WAN performance, failure recovery, and data consistency in an Active-Active NAS DR architecture.



Component	Role
NAS	Production-grade file storage environment
Linux Server	NFS mounting and real-time file change generation/validation
PeerGFS Agent	Real-time file change detection and Delta-Level Replication
Peer Management Center	Replication status monitoring, job management, and operational visibility
WAN Network	Simulation of latency, packet loss, bandwidth constraints, and unstable WAN conditions

Server Specifications
- CPU: 8 Core or higher - Memory: 32GB or more - Disk: SSD 50GB or more - NIC: 10Gbps - OS: Linux (Ubuntu/RHEL-based Linux OS)

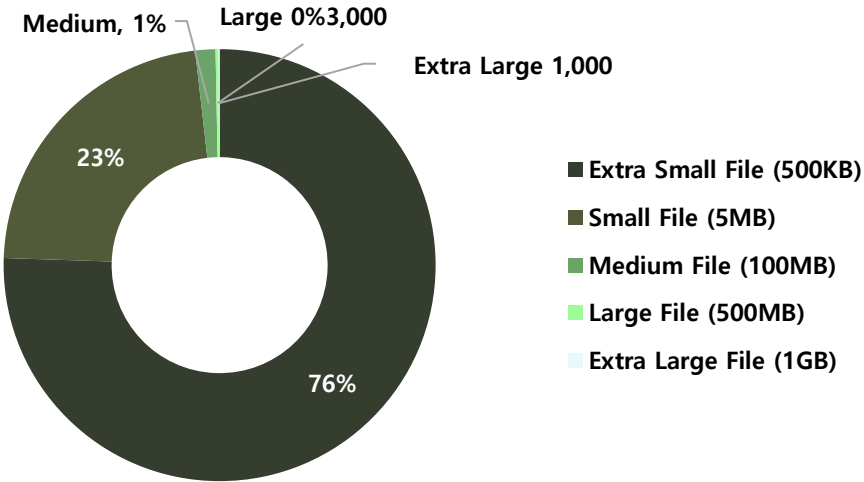
Notes
- SSD-based configuration for high-throughput replication performance validation - Linux environment optimized for large-scale file processing and replication validation 10Gbps NIC configuration for high-throughput WAN replication testing

Validation Dataset Design & Workload Configuration

The validation dataset is configured to reflect various file sizes and access patterns encountered in actual production environments. It is designed to validate multi-file processing workloads, large file replication, continuous file modifications, and real-time data synchronization characteristics within WAN environments. In addition, the validation is based on datasets reflecting actual NAS operational environments, including Office documents, images, and backup files.

File Composition (Total 1,324,000 files / Total 10TB)

File Types	Count	Scope
Extra Small File (500KB)	1,000,000	File and Directory Processing Performance, Concurrent File Changes, IOPS Load
Small File (5MB)	300,000	Standard Business File Synchronization Patterns
Medium File (100MB)	20,000	Throughput Verification
Large File (500MB)	3,000	WAN Bandwidth Utilization
Extra Large File (1GB)	1,000	Large File Transfer Integrity & Stability



Design Objective

Large Volume of Small Files	Performance Degradation during Bulk File Creation / Handling Concurrent Multi-file Updates / Processing Delays in Large Multi-file Environments
A Subset of Medium/Large Files	WAN Bandwidth Utilization / Throughput Stability in WAN Environments / Large-scale Transfer Reliability
Data Composition	99% Synthetic Data (fallocate-based) + 1% Actual Files (Sample Replication & Modification of PDF, JPG, TXT, etc.)

The verification environment consists of approximately 10TB of data and over 1.32 million files. To minimize processing bottlenecks and file system stress in large-scale environments, directory distribution and I/O optimization strategies have been implemented.

Core Principles

Data Composition	99% Synthetic Data + 1% Actual Files
Initial Data	- No initial network-based synchronization for the total 10TB of data - Physical migration (Seeding) used to replicate actual production environments - Real-time replication verification performed only for delta changes after initial setup
Directory Distribution Strategy	- Prevention of processing bottlenecks caused by single directory congestion - Maintenance of file system performance in large-scale small file environments - Workload optimization for parallel file creation and scanning

File System Optimization

Feature	Purpose
noatime	Reduces unnecessary file system write operations
nodiratime	Minimizes directory access overhead
Directory Sharding	Prevents concentrated file access bottlenecks
SSD-based Storage	Optimizes I/O for large-scale small file environments
Linux inotify	Supports real-time file change detection

Directory Structure

```
bash
/data
├── small/ (00~63, 64 Directories)
│   └── 500KB × 1,000,000
├── medium/ (00~31, 32 Directories)
│   └── 5MB × 300,000
├── large/ (00~15, 16 Directories)
│   └── 100MB × 20,000
└── xlarge/
    ├── 500MB × 3,000
    └── 1GB × 1,000
```

Large-scale verification data was generated using a synthetic approach to optimize creation time and storage efficiency. Small-file workloads were generated based on random data to emulate realistic large-scale file processing environments, while large-scale files utilized fallocate-based allocation for rapid generation and Delta-Level Replication validation.

Small File (500KB x 1,000,000)	Medium File (5MB x 300,000)	Large/XLarge File (500MB / 1GB)	Delta-Level Replication Validation Considerations	Strategies for Production-Grade Realism
Verification Objectives • Large-scale multi-file workload generation • Replication of large-scale small file environments • Verification of concurrent file creation and modification workloads Generation Methodology • Random data generation via /dev/urandom • Parallel creation using GNU parallel • Implementation of 64-directory sharding	Verification Objectives • Replication of standard business file operation patterns • Establishment of a mixed workload environment Generation Methodology • Random data-based generation • Implementation of 32-directory sharding • Parallel generation to accelerate dataset creation	Verification Objectives • Validation of WAN throughput • Validation of large-scale file transfer stability • Establishment of a baseline for Delta-Level Replication validation Generation Methodology • Rapid space allocation using fallocate • Applying entropy via partial random overwrites • Ensuring efficiency in generation time and storage utilization	• Avoid full re-generation of large-scale files • Workload configuration based on partial file overwrites • Data design structured to verify that only changed data (delta changes) are transmitted • Implementation of modification patterns that maintain original file sizes	• Reflecting file size distributions that emulate actual production environments • Inclusion of high-frequency multi-file workloads • Application of modification patterns based on partial overwrites • Inclusion of large-scale transfer workloads within WAN environments

Recognizing that synthetic data alone cannot fully replicate the file diversity and modification patterns of actual production environments, we have incorporated real-world business files (e.g., PDF, JPG, TXT, Office documents). By applying partial overwrite-based modification patterns, we have established a workload that reflects realistic enterprise NAS operational environments.



Real-World File Workload Configuration

Objectives

- Integration of actual business file patterns
- Inclusion of diverse binary structures
- Replication of application-driven file operation behaviors

Included File Types

PDF / JPG / TXT / Office document samples

Generation Methodology

- Sample file replication
- Modification via partial random overwrites
- Applying diverse file modification patterns



Partial Overwrite-Based Modification Strategy

Objectives

- Validation of Delta-Level Replication capabilities
- Verification of full-file retransmission prevention
- Simulation of real-world file modification patterns

Principles

- Maintenance of original file size
- Modification of selected portions of files only
- Application of overwrite-based delta changes



Post-Generation Verification Procedures

Verification Items	Objectives
Total Capacity Check	Verification of the target dataset (10TB)
File Count Verification	Validation of the workload scale
File Integrity Verification	Verification of file integrity
Sampling Verification	Confirmation of successful and accurate data generation

Data Integrity Verification

- Execution of sampling-based file integrity verification
- Validation of data health and identification of potential corruption
- Establishment of the baseline dataset for subsequent DR recovery scenarios

Linkage to Delta-Level Replication Validation

By implementing partial overwrite-based modification patterns on real-world business files, the environment is configured to validate whether only changed data (delta changes) are transmitted at a realistic workload level.

Performance Baseline		Key Verification Points
Performance Baseline	1. Baseline Measurement	Identify maximum system performance prior to PeerGFS implementation; establish benchmarks
	2. Initial Replication	Measure initial data loading performance and reference replication performance for select datasets
Near Real-Time Replication Validation	3. Near Real-Time Replication	Measure Replication Latency and End-to-End (E2E) time; validate Low RPO capabilities
	4. Delta-Level Replication	Verify that only changed data (delta changes) is transmitted
Active-Active Operations	5. Active-Active File Services	Ensure stable bi-directional operations and data consistency during concurrent multi-site workloads
	6. WAN Condition Test	Validate replication stability under degraded WAN conditions (high latency/packet loss)
Failure & Consistency	7. Failure Scenario	Confirm automatic re-replication and data consistency after restoration
	8. Recovery Scenario	Confirm automatic re-synchronization and data consistency after restoration

1. Baseline Performance Measurement & 2. Initial Replication Validation

In this stage, we measure the system’s baseline performance prior to Peer GFS implementation, establishing a benchmark for all subsequent validation results. To emulate production-grade environments, the initial large-scale dataset is established via physical seeding, allowing the validation to focus specifically on the replication performance of subsequent delta changes.

1. Baseline Performance Measurement

Measurement Area	Objective	Tool
Network Throughput	Identify maximum transmittable bandwidth	iperf3
Network Latency	Establish baseline for network latency	ping / iperf3
NAS IOPS	Establish benchmarks for small-file processing performance	fio
Storage Throughput	Measure large-scale I/O processing performance	fio
I/O Latency	Establish baseline for storage response times	fio

2. Initial Synchronization Validation

The core objective of this validation is to evaluate real-time replication and Delta-Level Replication performance for operational data changes, rather than the initial full-data replication.

Initial Replication Strategy

- **Physical Seeding:** Applied to the large-scale initial dataset (10TB)
- **Bypassing Network-Based Full Replication:** Initial full-data replication over the network is excluded to reflect real-world migration patterns
- **Focus on Delta-Level Synchronization:** Validation is centered on the near real-time replication performance of modified data following the initial baseline setup

The following scenario measures Replication Latency—the actual time elapsed between file modification and its successful replication to the DR site.

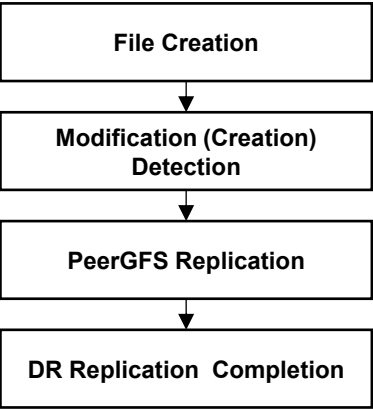
3. Near Real-Time Replication Validation

This scenario validates the delay between file modification events and replication completion at the DR site during continuous workload execution. The objective is to measure near real-time replication responsiveness and low-RPO operational capability.

Verification Objectives

- Measurement of actual replication latency between file modification and DR site replication.
- Validation of near real-time replication capabilities.
- Validation of Low RPO (Recovery Point Objective) environments.
- Verification of continuous near real-time replication stability under sustained enterprise NAS workloads.

Workflow



Measurement Metrics

Measurement Metrics	Description
Sync Detection Time	Time elapsed to detect file modifications
Transfer Start Delay	Latency before the data transfer process initiates
End-to-End Replication Time	Total time from initial file creation/modification to completion on the DR site
Sync Stability	Performance consistency and reliability under repetitive workload stress

Expected Outcomes

- Sub-second replication latency for standard enterprise NAS workloads.
- Confirmation of near real-time DR operational feasibility.
- Validation of sustained real-time replication without performance degradation.
- Successful verification of Low RPO environment requirements.

Test Procedures

1. Generate a file in the Main NAS environment.
2. Record the creation timestamp (START).
3. Record the timestamp when the file is first detected in the DR environment (DETECT).
4. Verify completion based on file size and data integrity validation.
5. Calculate the time difference (Latency) between the recorded timestamps.

Script Example (Bash)

```
bash

# 1. Generate file and record start time (Main Server)
START=$(date +%s.%N)
dd if=/dev/urandom of=/data/test/file_1 bs=1M count=10
echo $START > /tmp/start_time.txt

# 2. Record detection time on DR server
while [ ! -f /data/test/file_1 ]; do sleep 0.1; done
DETECT=$(date +%s.%N)

# 3. Verify sync completion via file size
while true; do
    size=$(stat -c%s /data/test/file_1)
    if [ "$size" -eq 10485760 ]; then break; fi
done
END=$(date +%s.%N)

# 4. Calculate results
echo "Sync Latency : $(echo "$DETECT - $START" | bc) second
s"
echo "E2E Total Time : $(echo "$END - $START" | bc) seconds
"
```

4. Delta-Level Replication Validation

This scenario evaluates the efficiency of PeerGFS Delta-Level Replication by verifying that only modified data is replicated instead of retransmitting entire files. The goal is to validate WAN bandwidth efficiency and replication performance optimization within enterprise NAS environments.

Verification Objectives

- Verify delta-level replication behavior: Confirm that only modified data is replicated.
- Validate minimization of network utilization: Ensure efficient use of bandwidth.
- Assess replication efficiency in WAN environments: Measure the effectiveness of delta-level replication over high-latency networks.
- Evaluate enterprise NAS workload optimization: Validate replication performance for large-scale file operations.

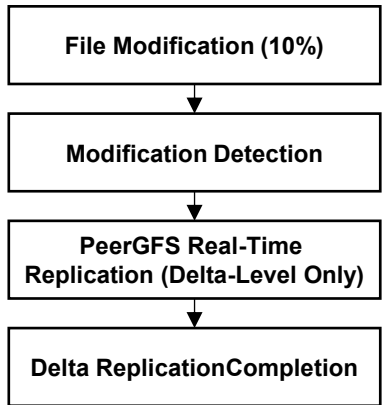
Test Principles

Principle	Description
Partial Overwrite	Modifying only a specific portion of the file
Maintain File Size	Prohibiting full-file regeneration to isolate the measurement of delta changes
Delta-Level Replication Validation	Confirming that only modified data is identified and replicated
Transfer Size Measurement	Conducting a comparative analysis of the actual data volume transmitted

Test Procedures

1. Generate a 100MB baseline file
2. Perform a 10MB (10%) partial overwrite on the file
3. Monitor the Peer Management Center to track real-time replication metrics
4. Conduct a comparative analysis of the actual transmission volume vs. total file size
5. Validate synchronization completion and data integrity on the DR site via checksum

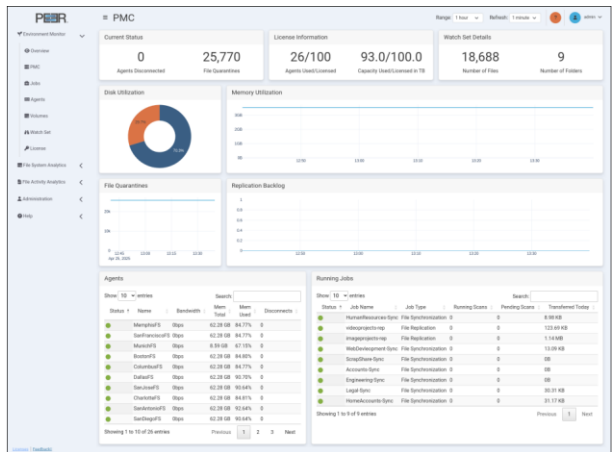
Workflow



Expected Validation Outcomes

- Successful replication of modified data only, without full-file retransmission
- Significant reduction in network utilization, particularly advantageous for WAN environments
- Confirmed operation of Delta-Level Replication: Validation that for a 100MB file modification, only the modified ~10MB data is replicated instead of retransmitting the full 100MB file

Verify via Peer Management Center



5. Active-Active File Services Validation

Traditional DR environments often suffer from low resource efficiency because the DR site remains in a passive operational state. This scenario validates the Active-Active File Services architecture, where both the Main and DR sites remain operational while maintaining data consistency and replication stability during concurrent bi-directional file operations.

Validation Objectives

- Validation of simultaneous operations at both Main and DR sites
- Verification of conflict-free bi-directional synchronization for file creation and modification
- Assessment of DR site feasibility for active production workloads
- Validation of sustained near real-time synchronization and data consistency

Test Configuration

Main Site	DR Site
File Creation/Modification	File Creation/Modification
Active	Active
↔ Bi-directional	↔ Bi-directional

Test Procedures

- Concurrent File Activities: Simultaneously create and modify files on both the Main and DR servers
- Cross-Site Modification: Perform separate file modifications on each site to induce bi-directional traffic
- Replication Monitoring: Verify the status and completion of bi-directional replication through the Peer Management Center
- Consistency Audit: Compare final file lists and data consistency between both sites

Measurement Metrics

Metric	Description
Bi-directional Replication	Success rate of simultaneous two-way Replication
Replication Latency	Time elapsed for modifications to be reflected across both sites
Data Consistency	Verification of final data consistency across both sites
Continuous Operation	Maintenance of stable Active-Active states under sustained load

Expected Outcomes

- Proven feasibility of bi-directional Active-Active operations, maximizing resource utilization at the DR site
- Validation of conflict-free, near real-time Replication during simultaneous multi-site updates
- Confirmed data integrity across all nodes, validating the robustness of the Peer GFS-based architecture

In real-world DR deployments, various WAN constraints such as latency, packet loss, and limited bandwidth are common. This scenario validates whether PeerGFS maintains real-time replication stability and WAN transfer efficiency even under degraded network conditions.

Verification Objectives

- Measure real-time replication performance changes under WAN conditions.
- Validation of response capabilities in high-latency scenarios.
- Assessment of stability in environments with packet loss.
- Verification of transmission efficiency under bandwidth-constrained conditions.

Measurement Metrics

Metric	Description
Transfer Speed	Fluctuations in throughput under WAN constraints
Replication Latency	Changes in end-to-end synchronization delay
Transferred Data	Total data volume transmitted via Delta Replication
Active Transfer	Monitoring of concurrent transfer threads and system status

WAN Simulation Phases

We utilize Linux tc (Traffic Control) to emulate realistic WAN conditions across the following phases.

Phase	Condition	Objective
Baseline	No constraints	Establish baseline performance benchmarks
Latency	50ms delay	Validate performance in high-latency (long-distance) environments
Packet Loss	1% loss	Verify retransmission efficiency and replication stability
Bandwidth Limit	100Mbps	Evaluate transfer efficiency under constrained bandwidth

Expected Outcomes

- Maintenance of stable synchronization even under severe WAN conditions
- Confirmed efficiency in bandwidth-limited environments, maximizing available throughput
- Validation of retransmission robustness in packet loss environments without session drops
- Proven feasibility for long-distance DR deployments through resilient transport protocols

This scenario evaluates the system's ability to maintain operations at the DR site during a Main site outage. The goal is to verify service continuity and confirm that the DR site can support production workloads independently without data loss.

Verification Objectives

- Validation of system response during a Main site failure
- Verification of DR site standalone operational capabilities
- Assessment of data loss during the failure event
- Validation of Business Continuity (BCP)

Test Procedures

1. Forced Shutdown: Execute a forced shutdown of the Main server
2. Standalone DR Operations: Continue production workloads exclusively on the DR server
3. Active File Modification: Perform file creation and modifications directly on the DR site
4. Service Audit: Verify service availability and identify any potential data loss

Verification Items

Item	Description
DR Continuous Operation	Feasibility of standalone DR operations
Data Loss	Presence of data loss at the point of failure
Sync Status	Synchronization state at the moment of failure
Service Availability	Ability to sustain services without interruption

Expected Outcomes

- Seamless operation at the DR site during Main site failure
- Zero or minimal data loss confirmed through PeerGFS real-time replication architecture
- Validation of Active-Active DR robustness for business continuity

This scenario validates the automatic restoration process once the Main site is recovered. It focuses on the system's ability to automatically re-synchronize changes made during the downtime and ensure perfect data integrity.

Verification Objectives

- Verification of automatic re-replication after Main site recovery
- Validation of delta change application for the period of failure
- Assessment of post-recovery data consistency
- Validation of stable operational failback

Test Procedures

1. Main Site Restoration: Bring the Main server back online
2. Auto Re-Replication Monitoring: Monitor PeerGFS for automatic replication resumption
3. Delta Reflection: Verify that modifications made on the DR site during downtime are applied to the Main site
4. Consistency Audit: Compare final data consistency between Main and DR sites

Verification Items

Item	Description
Auto Re-Synchronization	Capability to resume sync without manual intervention
Recovery Time	Time required to complete the re-synchronization process
Data Consistency	Post-recovery integrity check using checksums
Missing Changes	Verification that all changes from the downtime are reflected

Expected Outcomes

- Zero-touch automatic re-replication without manual intervention
- Successful reflection of all delta changes made during the outage
- Perfect data consistency maintained across both sites after recovery
- Stable and reliable operational failback confirmed

Summary of Validation Results & Expected Effects

This validation goes beyond simple file replication tests; it is designed to prove that a PeerGFS-based Active-Active NAS DR architecture is fully capable of stable, production-grade operations. The results comprehensively demonstrate the feasibility of real-time replication, WAN optimization, and multi-site data consistency.

Core Validation Summary

Validation Item	Key Findings & Verification
Near Real-Time Sync	Achievement of near real-time synchronization performance
Delta Replication	Maximized WAN efficiency through block-level delta transfers
Active-Active	Proven feasibility of bi-directional active operations at both sites
WAN Validation	Sustained synchronization stability under degraded network conditions
Failure & Recovery	Confirmed service continuity and automatic zero-touch recovery

Expected Effects

Operational Aspect

- Maximization of DR Site
Resource Utilization: Transitions the DR site from a passive to an Active-Active state, improving overall infrastructure efficiency
- Continuous Near Real-Time Replication DR Environment: Establishes a highly responsive disaster recovery architecture based on Low RPO

Technical Aspect

- WAN Environment Optimization: Overcomes network constraints PeerGFS replication architecture
- Significant Bandwidth Savings: Reduces network overhead via Delta-Level Replication
- Guaranteed Data Consistency: Ensures real-time data consistency across all sites

Business Aspect

- Minimized Failure Response Time: Achieves rapid recovery and minimizes the impact of system outages
- Enhanced Service Continuity: Ensures uninterrupted business operations through robust failover capabilities
- Reduced Operational Risk: Mitigates the risk of data loss and downtime in enterprise-scale environments